



Entamoeba histolytica — MCQ Practice

Microbiology GIT Block · MEDucation by Mattin Majid

HOW TO USE THIS SET

20 questions, 5 options each (A–E). Answer on paper or in your head, then check every answer against the key on the final page.

4 Easy · 10 Medium · 6 Hard · suggested time ≈ 25 minutes.

Q Questions

1. What are the two forms in the *E. histolytica* life cycle?

EASY

- A. Trophozoite and sporozoite
- B. Cyst and merozoite
- C. Oocyst and trophozoite
- D. Cercaria and metacercaria
- E. Cyst and trophozoite

2. What is the primary route of transmission for *Entamoeba histolytica*?

EASY

- A. Airborne droplets
- B. Fecal-oral route
- C. Mosquito bite
- D. Direct skin penetration
- E. Transplacental transmission

3. Which organ is most commonly affected in extraintestinal amebiasis?

EASY

- A. Liver (liver abscess)
- B. Spleen
- C. Kidney
- D. Pancreas
- E. Thyroid

4. Which form of *E. histolytica* is responsible for transmitting infection to a new host?

EASY

- A. Trophozoite
- B. Cyst
- C. Merozoite
- D. Gametocyte
- E. Oocyst

5. In which part of the gut does excystation of ingested *E. histolytica* cysts occur?

MEDIUM

- A. Stomach
- B. Duodenum
- C. Esophagus
- D. Rectum only
- E. Terminal ileum or colon

6. Which cytoplasmic feature of the *E. histolytica* trophozoite nucleus is characteristic on microscopy?

MEDIUM

- A. A large eccentric nucleolus with coarse chromatin clumps
- B. Multiple nuclei arranged peripherally
- C. Absence of any visible nuclear membrane
- D. Fine, regular peripheral chromatin with a small central karyosome
- E. A bilobed nucleus resembling *Giardia*

7. What is the role of amebapores in *E. histolytica* pathogenesis?

MEDIUM

- A. They mediate initial adherence to epithelial cells
- B. They are secreted antibodies that neutralize host immunity
- C. They encapsulate the trophozoite to evade phagocytosis
- D. They digest ingested bacteria within the food vacuole
- E. They form pores in lipid bilayers, causing cytolysis and inducing apoptosis

8. How do *E. histolytica* cysteine proteinases contribute to immune evasion?

MEDIUM

- A. By directly killing host CD8+ T cells
- B. By blocking cyst formation in competing *E. dispar*
- C. By cleaving and inactivating complement proteins C3a/C5a as well as IgA and IgG
- D. By preventing macrophage differentiation entirely
- E. By upregulating host interferon-gamma production

9. Which immune mechanism is primarily responsible for macrophage-mediated killing of *E. histolytica* trophozoites?

MEDIUM

- A. Direct antibody-mediated lysis without cell contact
- B. Contact-dependent killing involving oxidative pathways and nitric oxide, driven by interferon-based activation
- C. Complement-independent opsonization alone
- D. Mast cell degranulation
- E. Eosinophil major basic protein release

10. Why might stool examination be negative in a patient with amebic liver abscess?

MEDIUM

- A. Liver abscess never originates from intestinal infection
- B. *E. histolytica* cannot survive passage through the liver
- C. Stool examination findings in amebic liver abscess patients are usually negative, as the trophozoites have disseminated rather than remaining luminal
- D. Trophozoites are destroyed by bile before reaching the stool
- E. Stool antigen tests always cross-react with liver antigens

11. Which culture method is used for *E. histolytica*, and what is a key limitation?

MEDIUM

- A. Xenic cultivation on modified Locke-egg media; it is technically difficult and less sensitive than microscopy
- B. Blood agar culture; rapid but non-specific
- C. Viral cell culture lines; requires a BSL-4 lab
- D. Thioglycollate broth; always more sensitive than microscopy
- E. MacConkey agar; selects for *E. histolytica* specifically

12. Why is real-time PCR preferred over stool antigen detection for diagnosing *E. histolytica* where available?

MEDIUM

- A. PCR is faster to perform at the bedside
- B. Antigen tests cannot detect *E. histolytica* at all
- C. PCR does not require any laboratory equipment
- D. Antigen tests are more expensive than PCR in all settings
- E. PCR is considerably more sensitive than stool antigen detection

13. A patient with symptomatic intestinal *E. histolytica* infection has negative serology on initial presentation. What is the appropriate next step?

MEDIUM

- A. Discard serology as unreliable and rely on stool exam alone
- B. Conclude the patient does not have amebiasis and seek an alternative diagnosis
- C. Repeat serologic testing one week later
- D. Immediately start empiric antibiotics for bacterial dysentery
- E. Order a liver biopsy regardless of symptoms

14. Which assay is most commonly used worldwide to detect serum antilectin IgG antibodies against *E. histolytica*?

MEDIUM

- A. Western blot
- B. Complement fixation test
- C. ELISA
- D. Agglutination test only
- E. Flow cytometry

15. Amebic dysentery is most likely to be confused with which of the following based on clinical presentation?

HARD

- A. Viral gastroenteritis only
- B. Peptic ulcer disease
- C. Acute appendicitis
- D. Shigellosis or ulcerative colitis
- E. Cholecystitis

16. A patient with amebic dysentery has fewer stool leukocytes than expected for the severity of bloody diarrhea. What explains this finding?

HARD

- A. Trophozoite-mediated lysis of neutrophils reduces fecal leukocyte numbers
- B. *E. histolytica* never causes mucosal inflammation
- C. The patient must have a concurrent immunodeficiency
- D. Amebic dysentery never involves neutrophil recruitment
- E. Leukocytes are destroyed by the acidic colonic environment

17. Which statement best explains why *E. dispar* does not require public health intervention despite being found alongside *E. histolytica* in stool surveys?

HARD

- A. *E. dispar* is a nonpathogenic commensal, notably seen in HIV-positive patients, and does not cause invasive disease
- B. *E. dispar* cannot survive outside the human host
- C. *E. dispar* is eradicated by any antiparasitic therapy given for other infections
- D. *E. dispar* causes identical disease but is simply less common
- E. *E. dispar* cannot be transmitted via the fecal-oral route

18. A patient with acute invasive amebiasis shows a blunted T-cell response to *E. histolytica* antigens despite active disease. What is the proposed mechanism?

HARD

- A. The patient has an undiagnosed primary immunodeficiency
- B. *E. histolytica* antigens are non-immunogenic by nature
- C. Cysteine proteinases destroy all circulating T cells
- D. A parasite-induced serum factor depresses the T-cell response during acute invasive disease
- E. This is a universal, unrelated age-related finding

19. Why does sexual transmission of *E. histolytica* occur specifically in the context of oral-anal practices (anilingus)?

HARD

- A. The organism is sexually transmitted via genital secretions
- B. It creates a direct fecal-oral transmission route bypassing usual food/water vehicles
- C. *E. histolytica* specifically targets genital mucosa
- D. This transmission route requires an insect vector co-exposure
- E. Oral-anal contact inactivates the protective mucosal IgA response

20. In a patient with suspected extraintestinal amebiasis but negative initial imaging and serology, what diagnostic approach is specifically recommended by the lecture?

HARD

- A. Repeating the same imaging study daily for a week
- B. Proceeding directly to open surgical exploration
- C. Abandoning the diagnosis and treating empirically for malignancy
- D. A therapeutic trial with an amebicide
- E. Performing a bone marrow biopsy

✓ Answer key

Q	Answer	Difficulty	Why
1	E	EASY	<i>E. histolytica</i> exists as an infective, dormant cyst and an active, disease-causing trophozoite.
2	B	EASY	Amebiasis is transmitted via the fecal-oral route through contaminated food, water, or surfaces.
3	A	EASY	Liver abscess is the most common manifestation of extraintestinal amebiasis.
4	B	EASY	The cyst is the infectious, environmentally resistant form responsible for transmission.
5	E	MEDIUM	Excystation occurs in the terminal ileum or colon, releasing motile trophozoites.
6	D	MEDIUM	The nuclear membrane is lined by fine, regular chromatin granules with a small, centrally located karyosome (endosome).
7	E	MEDIUM	Amebapores are a family of peptides that form pores in lipid bilayers, contributing to cytolysis and apoptosis.
8	C	MEDIUM	Cysteine proteinases cleave and inactivate anaphylatoxins C3a and C5a, as well as IgA and IgG, aiding immune evasion.
9	B	MEDIUM	Lymphokines like interferon activate macrophages to kill trophozoites via contact-dependent, oxidative, nonoxidative, and nitric oxide pathways.
10	C	MEDIUM	Despite the hepatic origin from gut invasion, stool exam findings in amebic liver abscess are usually negative.
11	A	MEDIUM	Xenic cultivation using modified Locke-egg media has a 50-70% success rate but is technically difficult and overall less sensitive than microscopy.
12	E	MEDIUM	Real-time PCR offers considerably greater sensitivity compared to stool antigen detection methods.
13	C	MEDIUM	Serologic examination should be repeated one week later in those with a negative test on presentation, since antibody responses may take time to develop.
14	C	MEDIUM	ELISA is the most commonly used assay worldwide for detecting serum antilectin IgG antibodies.
15	D	HARD	Amebic dysentery can be confused with shigellosis, salmonellosis, schistosomiasis, or ulcerative colitis based on overlapping symptoms.
16	A	HARD	Amebic dysentery characteristically shows fewer fecal WBCs than bacterial dysentery because trophozoites lyse neutrophils at the site of invasion.
17	A	HARD	Molecular studies show <i>E. dispar</i> is a distinct, nonpathogenic commensal species (seen for example in HIV patients), unlike the pathogenic <i>E. histolytica</i> .
18	D	HARD	During acute invasive amebiasis, T-cell responses to <i>E. histolytica</i> antigens are depressed by a parasite-induced serum factor, a specific immune evasion mechanism.
19	B	HARD	Anilingus creates a direct fecal-oral exposure route, explaining why this specific sexual practice is a recognized transmission risk.
20	D	HARD	Extraintestinal infections can be diagnosed through imaging, serologic testing, or a therapeutic trial with an amebicide when other methods are inconclusive.